Media Tools and Integration

Computing and Mathematics

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Media Tools and Integration (A13952)

Short Title:	Media Tools and Integration		
Faculty	Science and Computing		
Department	Computing and Mathematics		
Cluster	Media Production		
Author	CDUNPHY		
Credits	10	**Level:**	Advanced

Description of Module / Aims

This module builds on the student's existing knowledge of audio/visual recording and editing and concentrates on software tool-specific features. The module also looks at bringing together content from multiple sources and integrating it in media-rich digital artefacts.

Programmes

`COMP-0670`	BSc (Hons) in Applied Computing (`WD_KCOMP_B`)	4 / 7 / E

Indicative Content

- Working with professional-grade audiovisual software tools and hardware
- Arranging, mixing and bouncing complex audio works
- Sound Design and synchronisation for video
- Production of audiovisual media for live performance and/or installation
- Troubleshooting and optimisation
- Lighting
- Projection
- Sample Project: Conceptualise, plan, create and deploy a digital mixed media installation. e.g. live music performance with video projection & lighting; A digital media mashup

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Propose and defend a concept for live performance or installation (Evaluation).
2. Assemble audio and visual media from multiple sources to create improved artefacts (Creating).
3. Plan and develop an audio/visual live performance or installation that contains an algorithmic element (Creating).
4. Solve simulated and real integration problems as they occur (Creating).

Learning and Teaching Methods

- This module will be characterised by student participation in both formal theoretical and practical/studio classroom activities in which students learn theoretical concepts and apply them immediately using dedicated hardware and software.
- Studio Classes will be delivered in two 2-hour blocks.
- Apple Mac Computer Lab with relevant hardware & software (kept up to date at least annually e.g. AdobeCC, iMovie, Resolume ARENA, Ableton Live, Traktor)
- Accessible, professional sound proofed recording studios with professional microphones and audio workstations (Vibe FM studios)
- DMX Lighting setup
- Access to cameras

- Video/Photography Studio
- Project Space Room
- Selection of MIDI controllers
- Sound System PA
- Data Projectors

Assessment Methods

	Weighing	Outcomes Assessed
Final Project	100%	1,2,3,4

Assessment Criteria

<40%: Unable to propose, conceptualise, plan, develop, create and integrate using digital mixed media tools.

40%-49%: Be able to propose, conceptualise, plan, develop, create and integrate using digital mixed media tools.

50%-59%: As above, incorporating creative approaches to artefacts and tool use.

60%-69%: As above and be able to problem-solve in a ``live'' setting.

70%-100%: All the above to an excellent level.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Practical	96	
Independent Learning	174	

Supplementary Material

- "Create Digital Music." http://createdigitalmusic.com/
- "DJ Tech Tools." http://www.djtechtools.com
- "lynda.com." www.lynda.com
- Claiborne, V. _Media Servers for Lighting Programmers_. Oxford: Focal Press, 2014.

Requested Resources

- CREATIVE AND PERFORMING: Special Purpose Area
- Room Type: Computer Lab